American Computer Science League

2024-2025 ● Contest 1: Short Problems Solutions ● Senior Division

1. Computer Number Systems

$9BD_{16} = 1001\ 1011\ 1101_2$
$= 100\ 110\ 111\ 101_2$
$= 4\quad 6\quad 7\quad 5_8$

C. 4675

2. Computer Number Systems

$A5C_{16} = 10 * 16\text{^}2 + 5 * 16\text{^}1 + 12 * 16\text{^}0$
$= 10 * 256 + 5 * 16 + 12 * 1$
$= 2560 + 80 + 12$
$= 2652$

D. 2652

3. Recursive Functions

$f(1) = 2$
$f(2) = f(2-1) - 1 = f(1) - 1 = 2 - 1 = 1$
$f(3) = f(3-1) + 3 = f(2) + 3 = 1 + 3 = 4$
$f(4) = f(4-1) - 1 = f(3) - 1 = 4 - 1 = 3$
$f(5) = f(5-1) + 3 = f(4) + 3 = 3 + 3 = 6$
$f(6) = f(6-1) - 1 = f(5) - 1 = 6 - 1 = 5$
The sequence of terms is: 2, 1, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11, 14, …
The pattern is subtract 1, add 3. So the 10th term is 9.

B. 9

4. Recursive Functions

$f(20,12) = f(12-2, 20-3) + 1 = f(10,17) + 1 = 17 + 1 = 18$
$f(10,17) = f(10+2, 17-2) - 2 = f(12,15) - 2 = 19 - 2 = 17$
$f(12,15) = f(15-2, 12-3) + 1 = f(13,9) + 1 = 18 + 1 = 19$
$f(13,9) = f(9-2, 13-3) + 1 = f(7,10) + 1 = 17 + 1 = 18$
$f(7,10) = 7 + 10 = 17$

C. 18

5. What Does This Program Do?

What is the output of the following program?

```
m = 369
b = int(m / 100)
m = m % 100
q = int(m / 25)
m = m % 25
d = int(m / 10)
m = m % 10
n = int(m / 5)
p = m % 5
output b, q, d, n, p
```

E. 3, 2, 1, 1, 4

6. What Does This Program Do?

In tracing the following program, the result of 2024 is the 22nd tetrahedral number. These numbers are also called triangular numbers.

```
t = 0
n = 1
while t != 2024
    s = 0
    for y = 1 to n
        s = s + y
    next y
    t = t + s
    n = n + 1
end while
output n-1
```

```
Each time the next value
to be added is one more
than the previous one.

1 + 3 + 6 + 10 + 15 + 21
+ 28 + 36 + 45 + 55 + 66
+ 78 + 91 + 105 + 120 +
136 + 153 + 171 + 190 +
210 + 231 + 253 = 2024

There are 22 of these.
```

D. 22